Lesson 4-3 · Period 2

Multiply & Divide Rational Expressions + DOK-3 Closure

Klimsara-adapted · Student-centered · No Blooket

Do Now — Simplify and state the domain

DOK 1 | 5 min | *bridge from P1*

Do Now · Simplify this rational expression What is the simplified form of the rational expression? What is the domain?

$$\frac{x^3 + 9x^2 - 10x}{x^3 - 9x^2 - 10x}$$

Factor every numerator and denominator completely. State all excluded values.

Launch — Example 3 (Multiply & Divide Rules)

DOK 2 | 12 min | *teacher models*

Multiply / Divide Rules (keep on your page)

1. Factor every numerator AND denominator completely.
2. For division, rewrite as multiplication by the **RECIPROCAL** of the divisor.
3. Cancel only common **FACTORS** across any numerator with any denominator.
4. State the domain: exclude every zero of every denominator that appeared in **ANY** step, including the reciprocal's new denominator.

Division rule: $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$ **Flip FIRST — then cancel.**

Explore — TPS + DOK-3 Performance Task

DOK 2-3 | 33 min | *student work*

7 items in order. Plan ~12 min for Practice #13 (Closure Argument).

Try It 3 · Multiply rational expressions Factor, cancel, state domain. (Two parts.)

Practice #27 · Multiply with factoring $\frac{(x-y)^2}{x+y} \cdot \frac{3x+3y}{x^2-y^2}$ — find product and domain.

Try It 4 · Divide rational expressions Flip the divisor first. State domain including reciprocal's denominator.

Share / Summary

DOK 2 | 5 min | *academic conversation + P3 bridge*

Sentence frame · Closure Argument "I rewrite the division as multiplication by the reciprocal, giving ______. After factoring, I cancel the factor ______, leaving ______. The domain excludes $x =$ ______ because those values make at least one denominator zero at some step."

Bridge to Period 3 Next period we apply multiply/divide skills to complex rational expressions and assessment-critical items. Bring your domain-restriction list habit.

Exit Ticket — Summary Recap

Algebra 2 · Lesson 4-3 · Period 2

DOK 1-2 **3 min** *not graded*

Complete on your exit ticket

1. To divide rational expressions I first ____________________________.
2. I cancel common factors, not ____________________________.
3. My domain restriction list must include zeros from ____________________________.
4. One biggest thing I learned today: ____________________________.